

PAPER_I_EXT_AUDIT — README

PAPER_I_EXT_AUDIT_2026-07-10 — External Adversarial Audit of Paper I

Independent external audit (adversarial-math-audit protocol) of Theorem 1.1 of

Affine Profile Reduction for Fractional Triangle Packings in Split Graphs:

> For every split graph G on n vertices, $|E(G)| - 2 \cdot v^*(G) \leq n^2/6 + n$.

Verdict: **PASS** (intermediate AI-assurance tier — not human peer review).

Contents

- `FINAL_AUDIT_REPORT.md / .pdf` — executive verdict, gate table (A–J), computational audit, falsification log, coverage declaration, recommendation.
- `ENVIRONMENT.md / .pdf` — independent environment + reproducibility record.
- `scripts/` — four self-contained scripts, each with its `.py`, `results.txt`, English `report.md/.pdf`, a `.zip` of the folder, and `.zip.sha256`:
 - `01_algebraic_identities` — exact symbolic check of every ledger identity (Gate E).
 - `02_orbit_lp_and_reduction` — orbit LP = closed form, orbit symmetrization lossless, affine reduction + uniform bound (Gates B, C).
 - `03_split_falsification` — v^* LP falsification on 622 split graphs + extremal tightness (Gate F).
 - `04_lean_certificate` — independent Lean rebuild + `#print axioms` + SHA + statement match (Gate H); includes `independent_build.log`.
- `_lib/` — `pdfgen.py` (PDF renderer), `package.py` (packager).
- `SHA256_MANIFEST.txt` — digests of all package files.

Result summary

Gate	Result
A definitions consistent	VERIFIED
B affine reduction	VERIFIED
C orbit symmetrization + finite LP	VERIFIED
D finite LP duality in-hypotheses	VERIFIED
E deficit identity + $R(p,q)$ constant	VERIFIED (9/9 identities exact)
F conversion to n + tightness	VERIFIED (0/622 violations; extremal attains $n^2/6+n/6$)
G assembly, no circularity	VERIFIED

Gate	Result
H Lean certificate	VERIFIED (indep. build 8027 jobs, axioms clean, SHA match)
I citations	VERIFIED_WITH_RESIDUAL_RISK (duality standard + Lean-derived)
J scope / no over-claim	VERIFIED

No counterexample, circularity, or over-claim found. Human/journal peer review remains the appropriate next tier.

Reproduce

```
cd scripts/01_algebraic_identities && python 01_algebraic_identities.py
cd ../02_orbit_lp_and_reduction    && python 02_orbit_lp_and_reduction.py
cd ../03_split_falsification       && python 03_split_falsification.py
# Gate H (independent Lean build): copy 05_formalization/lean sources to a SHORT path,
#   enable git core.longpaths, `lake exe cache get`, `lake build PaperI.PaperI_Statement`,
#   then: cd scripts/04_lean_certificate && python 04_lean_certificate.py
python _lib/package.py             # render PDFs, zip + SHA, manifest
```

Requires Python 3.14 (sympy, scipy/HiGHS, numpy, reportlab); Lean v4.28.0 + Mathlib v4.28.0 for Gate H. Master seed 20260710.